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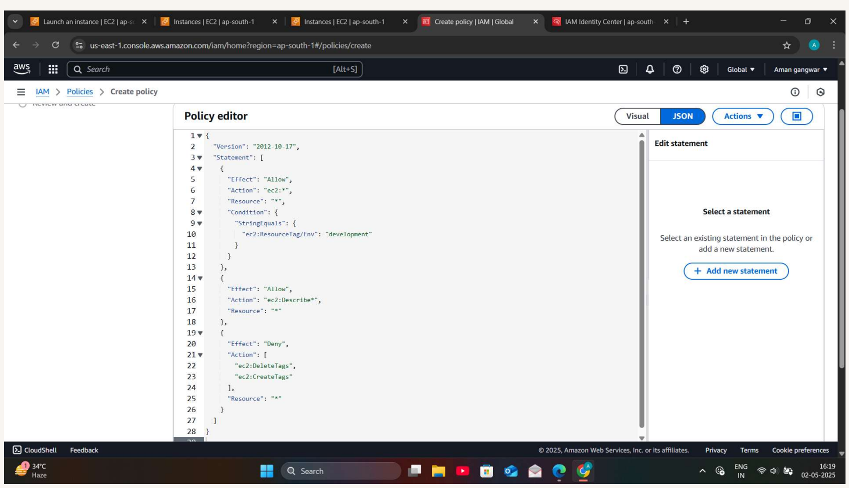
Cloud Security with AWS IAM

Aman gangwar



Introducing Today's Project!

In this project, we will demonstrate how to use AWS IAM to control access and permission setting in your account.We're doing this project to learn about cloud security from the absolute foundations :)

Tools and concepts

Services we used today were EC2 and IAM! Key concepts we learnt include IAM users,policies,user groups and account aliases.

Project reflection

This project took us approximately 1.5 hours today !The most challenging part was understanding the IAM policy since it was written in JSON and it contained multiple statements.

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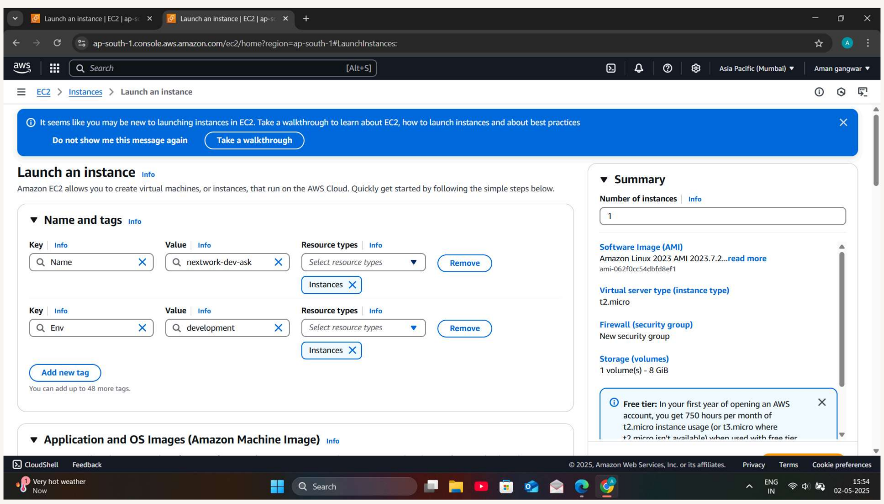
Tags

Tags are organisational tools that lets label our resources.They are helpful for grouping resources,cost allocation and applying policies for all resources with the same tag.

The tag I've used on my EC2 instances is called environment. The value I've assigned for our instances are production and development!



IAM Policies

IAM Policies are like rules that determine who can do what in our AWS account.We're using policies today to control who has access to our production/environment instance

The policy I set up

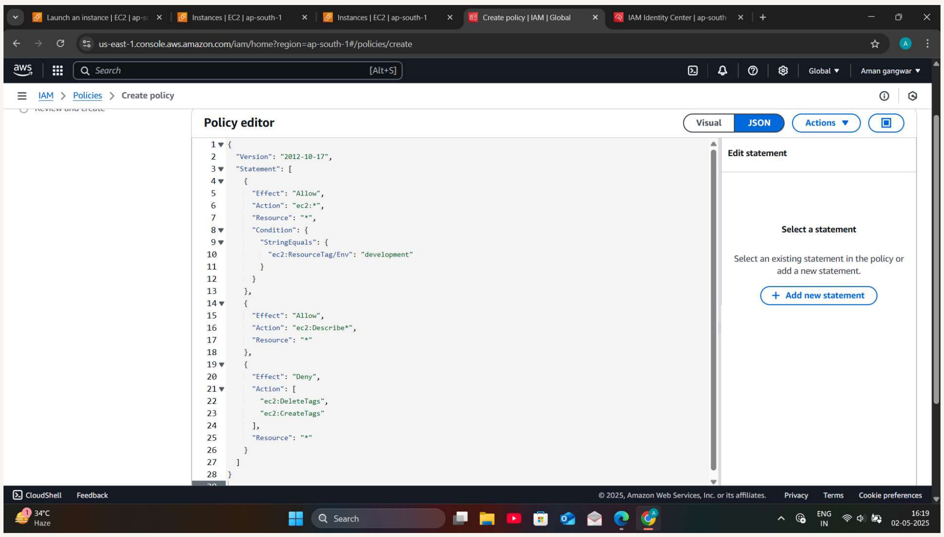
For this project, we've set up a policy using the JSON

We've created a policy that allows the policy holder to have permission to do anything they want to any instance tagged with "development".They can also see information, but they're denied access to deleting/creating tags for any instance as well.

When creating a JSON policy, you have to define its Effect, Action and Resource.

The Effect, Action, and Resource attributes of a JSON policy means whether or not the policy is allowing/denying action(i.e. Effect);what the policy holder can or cannot do and the specific AWS resources that the policy relates to(i.e. resource).

My JSON Policy



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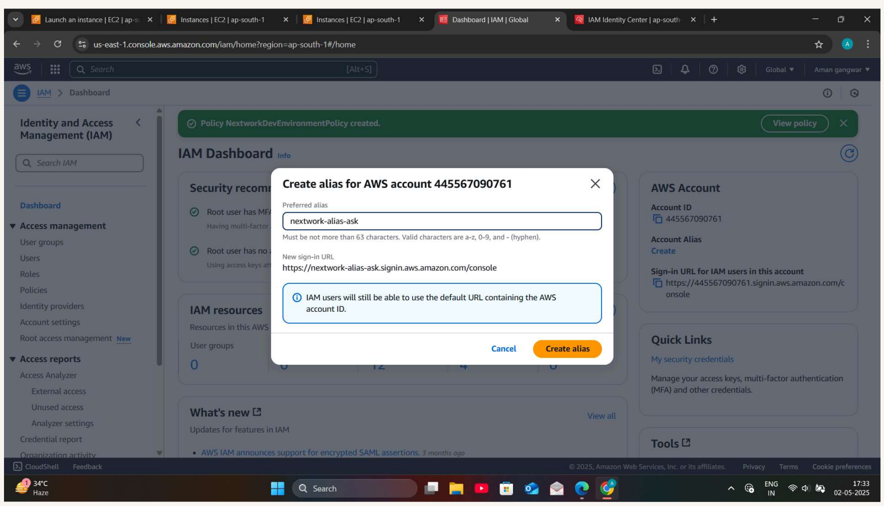
Account Alias

An account alias is simply a nickname for our AWS account!Instead of a long account ID, we can now reference our Account Alias instead!

Creating an account alias took us 30 seconds - it's a simple configuration in IAM dashboard. Now, our new AWS console sign-in URL is https://nextwork-alias-ask.signin.aws.amazon.com/console



IAM Users and User Groups

Users

IAM users are people or entities that have access/can login to our AWS account.

User Groups

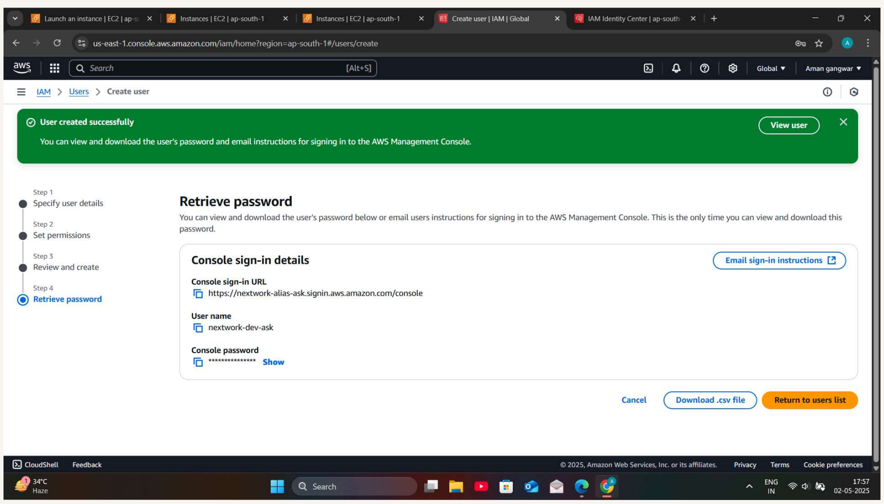
IAM user groups are like folders that collect IAM users so that you can apply permission setting at the group level.

We attached the policy I created to this user group, which means any user created inside this group will automatically get the permission attached to our NextworkDevEnvironmentPolicy IAM policy.

Logging in as an IAM User

The first way is to email sign-in instructions to the user, while the second way is to download the .csv file with the sign in details inside.

Once we logged in as my IAM user, we noticed that our user is already denied access to panels on the main AWS console dashboard, because we only set up permission to our development EC2 instance,so our intern wouldn't have access to see anything else



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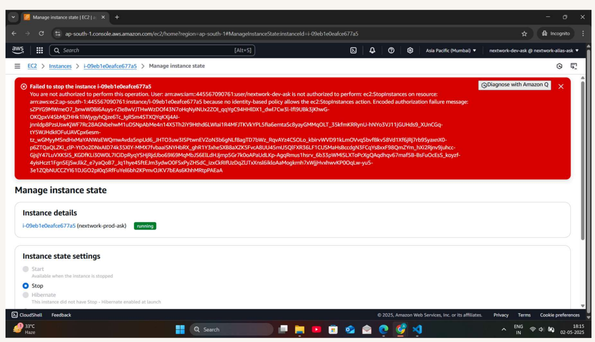
Testing IAM Policies

I tested my JSON IAM policy by attempting to stop both the development and the production instance.

Stopping the production instance

When we tried to stop it,we were met with an error! This was because our production instance is tagged with the 'production' label which is outside the scope of our permission policy-interns are only allowed to do things to development instances.



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Testing IAM Policies

Stopping the development instance

Next, when we tried to stop the development instance,we successfully saw the instance state change to stopping and then Stopped This was because our permission policy allows interns to stop instances.